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Closed-Loop feedback control system for EXOhSPEC

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ABSTRACT

EXOhSPEC is an ambient-air, fibre-fed, high-resolution spectrograph testbed in which optical-path-length metrology, image-centroid measurements and environmental telemetry are recorded during operation.

We evaluate a closed-loop architecture in which a thermo-electric cooler supplies slow image-position correction and a discrete fine-optics stage supplies bounded residual correction, and isolate each actuator's contribution against a matched, same-hardware open-loop baseline.

A 31.44-hour hybrid run sustained bounded operation over 1604 feedback frames, with axis-resolved scatter of 0.0635 pixels horizontally and 0.1524 pixels vertically about the operating reference and the radial image displacement remaining within 0.5 pixels throughout.

Disabling the fine-optics stage and holding the image with the thermal channel alone degraded stability to 0.316 pixels; disabling the thermal channel after warm-up and holding the image with the fine-optics stage alone matched the full hybrid result, 0.138 pixels, using an order of magnitude fewer corrective actions – indicating that fine-scale stabilisation is carried disproportionately by the fast actuator rather than the slow one.

A genuine open-loop baseline, recorded on the same hardware immediately after an extended hybrid run, showed an eleven-fold increase in image-position scatter once all actuation was disabled: the first directly measured, rather than inferred, quantification of the closed loop's benefit in this programme.

The independently monitored optical path continued to evolve throughout every closed-loop record, showing that a controlled image position is a distinct outcome from a stationary optical path.

The control law, its identified environmental model and its complete safety logic are reported in full, giving a reproducible, guarded architecture whose actuator roles are now supported directly by a matched open-loop and single-actuator comparison rather than by inference.

Key words: instrumentation: spectrographs – techniques: radial velocities – optical metrology – control systems – atmospheric effects – methods: data analysis

1 INTRODUCTION

High-resolution radial-velocity spectrographs are stabilised as complete systems rather than by controlling a single temperature sensor or optical component.

The measured spectrum responds to the thermal and mechanical state of the optical train, to pressure and refractivity along the beam, and to detector behaviour, illumination stability, wavelength calibration and extraction (Fischer et al.

Since the first detections of exoplanets by the Doppler method (Mayor et al.

2003), the field has advanced by identifying each of these contributions and controlling it to a stated tolerance: HARPS and, more recently, ESPRESSO reach sub- m s^{-1} stability by combining vacuum enclosures, active thermal control and pressure regulation with careful fibre scrambling and calibration (Mayor et al.

2003; Pepe et al.

2021; Spronck et al.

The literature therefore separates design requirements, subsystem validation and performance verification, and this separation frames the present work: our data measure image-centroid displacement and one monitored interferometric path, not a complete stellar radial-velocity error budget.

The relevant instrumentation literature can be read as a sequence of methodological lessons rather than as performance benchmarks

that an ambient-air testbed should match.

The original EXOspec instrument established a compact, off-the-shelf, fibre-fed spectrograph with $R > 70,000$ in which thorium–argon line positions were monitored in real time and a piezo-actuated element removed image drift (Jones et al.

2021); the present paper does not repeat that optical-design story but asks the narrower question of whether optical-path monitoring, environmental telemetry, thermal correction and bounded fine-optics trim together form an interpretable closed loop, and whether each actuator's contribution can be isolated and measured directly.

The NEID environment-control programme shows how a precision-radial-velocity instrument converts an error-budget requirement into a documented control architecture and a long-duration stability demonstration in vacuum (Robertson et al.

2019); the implication for our work is that measured channels, time window, reference origin and actuator limits must all be stated, and that ambient-air control must not be presented as equivalent to vacuum stabilisation.

EXPRES verifies performance by decomposing the instrumental error budget into environmental, mechanical, optical, calibration, fibre and detector contributions (Blackman et al.

2020), which is why every claim here is restricted to the directly measured quantity and no radial-velocity number is inferred without a wavelength solution and extraction analysis.

Finally, the NEID fibre-feed study shows that a stable environmental trace alone

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does not eliminate illumination-driven error, so that image position, optical path, actuator commands and environmental context must be retained together rather than in isolation (Kanodia et al.

These requirements have been consolidated into comprehensive error budgets for next-generation Doppler spectrometers (Halverson et al.

2016), against which even laboratory subsystem tests can be positioned.

The physical motivation for recording pressure, temperature and humidity is specific and quantitative.

For a geometric path length L in air of refractive index n , the leading-order change in optical path is $\delta \text{OPL} \approx n \delta L + L \delta n$: a mechanical term and a refractivity term of comparable importance.

The refractivity of air, $n - 1 \approx 2.7 \times 10^{-4}$ at optical wavelengths, varies at the level of roughly 1×10^{-6} per kelvin, 3×10^{-7} per hectopascal and 1×10^{-8} per per-cent relative humidity near room conditions (Ciddor 1996; Bönsch & Potulski 1998).

Over a metre-scale air path a one-kelvin drift therefore changes the optical path by order a micron, which is directly comparable to the interferometric residuals reported below and to the pixel-scale image motion of interest.

Recording the atmospheric state synchronously with the optical path is thus not incidental telemetry but a prerequisite for interpreting the measured lengths at all; the same first-order relation, however, prevents a single centroid displacement from being assigned uniquely to a geometric or a refractive cause, because mechanical, thermal-boundary and illumination effects coexist.

Figure?

shows this coupling directly rather than only from the scaling argument above: a small step in the recorded temperature coincides with a distinct dip in the simultaneously monitored optical path, visible in both the raw and smoothed traces, over a 2-hour window in which humidity and pressure evolve smoothly and cannot account for the feature.

A closed-loop, or feedback, control system is one in which the quantity to be stabilised is measured, compared against a desired reference, and used to compute a corrective action that is applied back to the system, so that the effect of the correction is itself measured on the next cycle (Åström & Murray 2008).

This differs fundamentally from open-loop or feed-forward operation, in which a correction is applied from a model or a schedule without measuring its result: an open-loop scheme cannot compensate for a disturbance it did not anticipate, nor detect when its own model has ceased to be valid.

For a spectrograph the distinction is decisive.

The disturbances that move the image on the detector—laboratory temperature cycling, barometric drift, slow mechanical relaxation—are neither fully predictable nor constant between nights, so a fixed corrective schedule computed once cannot hold the image in place indefinitely.

A feedback loop instead measures the image position continuously and corrects only what it observes, which allows it both to reject unforeseen drift and, critically, to expose the moment its internal model no longer describes the instrument.

Implementing such a loop for EXOHSEC requires three elements: an observable that reports the controlled quantity, one or more actuators that can move it, and a decision layer that maps the measured error to a bounded actuator command.

Here the observable is the detector-plane centroid of a calibration feature, monitored alongside an independent optical-path length and the ambient state; the actuators are a slow thermoelectric channel that sets the enclosure thermal state and a fast fine-optics stage that trims the image position in discrete steps; and the decision layer identifies an environmental model from the current record and issues gated, magnitude-limited corrections. Sections 2 and 3 specify this architecture and its control law in full, including the calibration mathematics for the fine-optics stage.

Section 4 then reports, against recorded data, whether the resulting loop holds the image position within a controlled region, and – the

Figure 1.

Environmental coupling to optical path length in precision laser interferometry.

(a) Optical path length measurements over a 2-hour observation window showing raw data (light blue points) and smoothed trend (dark blue line, 150-sample moving average).

Red shaded region highlights a temperature perturbation event at $t \approx 0.3$ hours causing a visible optical path excursion.

(b) Temperature variation showing a dip of Δ

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.